

Test 10 — Answer Key

Each row gives the item number, the B.E.S.T. benchmark and difficulty, the correct answer, and a student-facing explanation.

#	Benchmark	Answer	Explanation
1	MA.6.NSO.3.3 moderate	$10^2 \rightarrow 100$; $10^3 \rightarrow 1000$; $2^3 \rightarrow 8$	The exponent tells how many times the base is used as a factor.
2	MA.6.GR.2.4 challenging	80	Add the areas of all six rectangles in the net; opposite faces match.
3	MA.6.NSO.3.5 moderate	$\frac{1}{2} \rightarrow 50\%$; $0.6 \rightarrow 60\%$; $\frac{1}{4} \rightarrow 25\%$	Write each number as a decimal and multiply by 100 to get its percent.
4	MA.6.NSO.1.1 easy	$-1 \frac{3}{4}$	$-1 \frac{3}{4}$ is more than -2 and less than -1 , so it sits between them on the number line.
5	MA.6.AR.3.1 challenging	5:3	Write bus students first, then divide both numbers by their greatest common factor.
6	MA.6.GR.2.3 challenging	50	Volume is length times width times height, or the area of the base times the height.
7	MA.6.DP.1.3 challenging	14	The median is shown by the vertical line inside the box.
8	MA.6.NSO.4.1 moderate	-10.0	Combine the two integers, then find that value on the number line.
9	MA.6.AR.3.4 moderate	64, 40	Write each percent as a fraction of 100 and multiply by the total.
10	MA.6.AR.3.3 moderate	6, 15	Multiply both numbers in a ratio by the same factor to keep it equivalent.
11	MA.6.AR.1.3 challenging	15	Evaluate the exponent before multiplying, then subtract.
12	MA.6.AR.1.2 challenging	$x \leq 48$	"At least" and "no more than" include the boundary value; "more than" and "under" do not.
13	MA.6.AR.2.3 challenging	5	Divide both sides by the coefficient to undo the multiplication.
14	MA.6.AR.1.1 challenging	$x - 9$	Translate each word phrase in order, keeping subtraction and division in the order they are written.
15	MA.6.GR.1.3 challenging	5	Points with the same y-coordinate lie on a horizontal line, so subtract the x-coordinates.
16	MA.6.NSO.3.1 easy	36	The next time they leave together is the least common multiple of the two intervals.
17	MA.6.NSO.3.2 easy	16	Add inside the parentheses first, then multiply by the factor outside.
18	MA.6.NSO.1.3 easy	$3 \frac{3}{4}$	The distance from 0 is the absolute value of the number.

#	Benchmark	Answer	Explanation
19	MA.6.AR.2.4 challenging	7.78	Subtract the known addend from the total.
20	MA.6.DP.1.2 challenging	23	Add all the values, then divide by how many values there are.
21	MA.6.DP.1.1 challenging	How many minutes did each sixth grader spend on the bus this morning?	A statistical question expects different answers from different people, so it produces a set of numerical data.
22	MA.6.AR.3.5 moderate	2 pounds, in ounces → 32; 3 gallons, in quarts → 12; 5 meters, in centimeters → 500	Multiply by the number of smaller units in one larger unit.
23	MA.6.NSO.2.3 easy	\$6.48	Find the cost of all the items first, then subtract that from the amount paid.
24	MA.6.NSO.3.3 easy	4/25	Square the numerator and square the denominator.
25	MA.6.GR.1.1 challenging	(-4, 3)	The first number tells how far to move right or left, and the second tells how far to move up or down.
26	MA.6.AR.1.4 challenging	$4x + 40$	Multiply the number outside the parentheses by each term inside.
27	MA.6.DP.1.6 moderate	Mean → $18 \frac{2}{3}$; Median → 18; Range → 23	Recompute each measure using the six values of the new data set.
28	MA.6.NSO.1.2 easy	$5 \frac{3}{4}$	The opposite of a number is the same distance from 0 but on the other side.
29	MA.6.DP.1.5 moderate	0	Count the values from the list that fall inside that interval.
30	MA.6.NSO.4.2 moderate	48, -6	Same signs give a positive result; different signs give a negative result.
31	MA.6.NSO.2.1 easy	5	Multiply both numbers by the same power of 10 so the divisor is a whole number, then divide.
32	MA.6.AR.2.1 challenging	3	Substitute each value for x and check which one makes both sides equal.
33	MA.6.AR.2.2 challenging	27	Subtract the same number from both sides to undo the addition.
34	MA.6.GR.2.2 challenging	150	The area of a parallelogram is base times height, using the height that is perpendicular to the base.
35	MA.6.NSO.3.4 moderate	2, 3, 7	A prime factor divides the number and has exactly two factors.
36	MA.6.NSO.1.4 easy	6	Take each absolute value first, then subtract.
37	MA.6.GR.2.1 challenging	42	A triangle is half of a rectangle with the same base and height, so use one half times base times height.

#	Benchmark	Answer	Explanation
38	MA.6.NSO.2.2 easy	3 1/5	Write the mixed number as a fraction, then multiply by the reciprocal of the divisor.